

Homework Set # 7

S5.3.1 $V(\vec{r}) = V_r(\vec{r}) - iV_i$

$$H = \frac{1}{2m} p^2 + V_r(\vec{r}) - iV_i \neq H^\dagger$$

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + [V_r(\vec{r}) - iV_i] \psi$$

$$-i\hbar \frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi^* + [V_r(\vec{r}) + iV_i] \psi^*$$

$$\frac{\partial}{\partial t} |\psi|^2 = \psi^* \frac{\partial \psi}{\partial t} + \frac{\partial \psi^*}{\partial t} \psi$$

$$= \frac{1}{i\hbar} \left[-\frac{\hbar^2}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) - 2iV_i |\psi|^2 \right]$$

$$= \frac{i\hbar}{2m} \vec{\nabla} \cdot (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*)$$

$$- \frac{2}{\hbar} V_i |\psi|^2$$

$$\frac{\partial}{\partial t} [|\psi|^2 e^{\frac{2}{\hbar} V_i t}] + \vec{\nabla} \cdot \vec{j} = 0$$

$$\vec{j} = \frac{\hbar}{2mi} (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*)$$

[Note If $\psi'(\vec{r}, t)$ is solution of

$$i \hbar \partial_t \psi' = - \frac{\hbar^2}{2m} \nabla^2 \psi' + V_r(\vec{r}) \psi'$$

then

$$\psi(\vec{r}, t) = e^{-\frac{i}{\hbar} V_i t} \psi'(\vec{r}, t) \quad]$$

$$p = \int d^3r |\psi(\vec{r}, t)|^2$$

$$\frac{dp}{dt} = - \frac{2}{\hbar} V_i p$$

$$p(t) = p(0) e^{-\frac{2}{\hbar} V_i t}$$

55.3.2

$$\psi = \xi \tilde{\psi}$$

ψ' real

$\xi = \text{complex number}$

$$\vec{j} = \frac{\hbar}{2mi} (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*)$$

$$= \frac{\hbar}{2mi} |\xi|^2 (\tilde{\psi} \vec{\nabla} \tilde{\psi} - \tilde{\psi} \vec{\nabla} \tilde{\psi}) = 0$$

S5.3.3

$$\psi_{\vec{p}} = \frac{1}{(2\pi\hbar)^{3/2}} e^{i/\hbar \vec{p} \cdot \vec{r}}$$

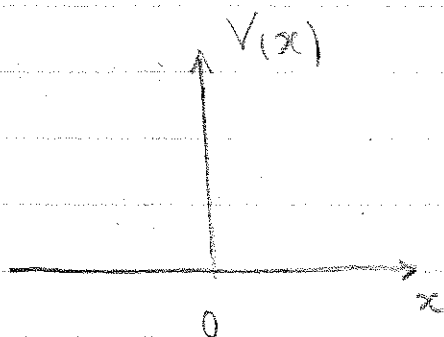
$$\vec{j} = \frac{\hbar}{2mi} (\psi_{\vec{p}}^* \vec{\nabla} \psi_{\vec{p}} - \psi_{\vec{p}} \vec{\nabla} \psi_{\vec{p}}^*)$$

$$= \frac{\hbar}{2mi} \left(\frac{1}{\hbar} \vec{p} \right) 2 \frac{1}{(2\pi\hbar)^3} = \frac{\vec{p}}{m} \frac{1}{(2\pi\hbar)^3}$$

$$\rho = \psi_{\vec{p}}^* \psi_{\vec{p}} = \frac{1}{(2\pi\hbar)^3}$$

$$\vec{j} = \vec{v} \rho \quad \text{with} \quad \vec{v} = \frac{\vec{p}}{m}$$

S5.4.2 a



$$V(x) = a V_0 \delta(x)$$

$$- \frac{\hbar^2}{2m} \left[\frac{d\psi}{dx}(0_+) - \frac{d\psi}{dx}(0_-) \right]$$

$$= \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} dx (E - V(x)) \psi(x)$$

$$= -a V_0 \psi(0)$$

$$x < 0 \quad \psi(x) = A_+ e^{ikx} + A_- e^{-ikx}$$

$$x > 0 \quad \psi(x) = B e^{ikx}$$

$$k = \frac{1}{\hbar} \sqrt{2mE}$$

$$\begin{cases} A_+ + A_- = B \\ + \frac{\hbar^2}{2m} ik [B - (A_+ - A_-)] = +a V_0 B \end{cases}$$

$$A_+ + A_- = B$$

$$A_+ - A_- = B \left(1 - \frac{2maV_0}{ik\hbar^2} \right)$$

$$A_+ = B \left(1 - \frac{m a V_0}{i k \hbar^2} \right)$$

$$A_- = B \frac{m a V_0}{i k \hbar^2}$$

$$R = \frac{|A_-|^2}{|A_+|^2} = \frac{\left(\frac{m a V_0}{k \hbar^2} \right)^2}{1 + \left(\frac{m a V_0}{k \hbar^2} \right)^2}$$

$$T = \frac{|B|^2}{|A_+|^2} = \frac{1}{1 + \left(\frac{m a V_0}{k \hbar^2} \right)^2}$$

S 5.3.4

$$\psi = A e^{\frac{i}{\hbar} p x} + B e^{-\frac{i}{\hbar} p x}$$

$$j = \frac{\hbar}{2mi} \left(\psi^* \frac{d\psi}{dx} - \psi \frac{d\psi^*}{dx} \right)$$

$$= \frac{\hbar}{2mi} \left[(A^* e^{-\frac{i}{\hbar} p x} + B^* e^{\frac{i}{\hbar} p x}) \times \right.$$

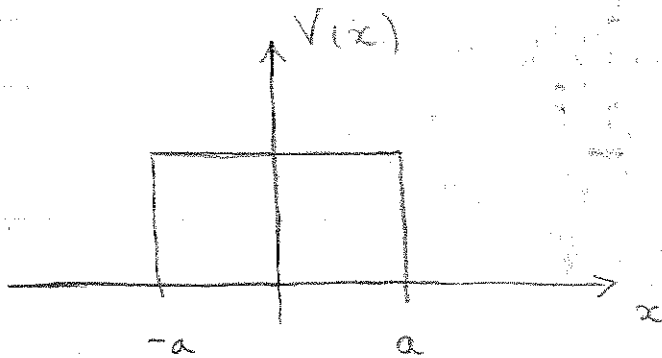
$$\left. \times \frac{ip}{\hbar} (A e^{\frac{i}{\hbar} p x} - B e^{-\frac{i}{\hbar} p x}) - \text{h.c.} \right]$$

$$= \frac{\hbar}{2mi} \left\{ [|A|^2 - |B|^2] + \underbrace{(B^* A e^{\frac{2i}{\hbar} p x} - \text{h.c.})}_{\text{pure imaginary}} \right\}$$

$$= \frac{\hbar}{2mi} \frac{ip}{\hbar} 2 (|A|^2 - |B|^2)$$

$$= \frac{p}{m} (|A|^2 - |B|^2)$$

b.



$$V(x) = 0 \quad \text{for } |x| > a$$
$$V(x) = V_0 \quad \text{for } |x| < a$$

$$E < V_0$$

$$x > a \quad \psi_E(x) = C e^{ikx}$$

$$|x| < a \quad \psi_E(x) = B_+ e^{+Kx} + B_- e^{-Kx}$$

$$x < -a \quad \psi_E(x) = A_+ e^{ikx} + A_- e^{-ikx}$$

$$k = \frac{1}{\hbar} \sqrt{2mE}$$

$$K = \frac{1}{\hbar} \sqrt{2m(V_0 - E)}$$

$$C e^{ika} = B_+ e^{Ka} + B_- e^{-Ka}$$

$$ik C e^{ika} = K (B_+ e^{Ka} - B_- e^{-Ka})$$

$$A_+ e^{-ika} + A_- e^{ika} = B_+ e^{-Ka} + B_- e^{Ka}$$

$$ik (A_+ e^{-ika} - A_- e^{ika}) = K (B_+ e^{-Ka} - B_- e^{Ka})$$

$$B_{\pm} = \frac{e^{\mp ka}}{2} \left(1 \pm \frac{ik}{k} \right) e^{ika} C$$

$$A_{\pm} = \frac{e^{ika}}{2} \left[\left(1 \mp \frac{ik}{k} \right) B_{+} e^{-ka} + \left(1 \pm \frac{ik}{k} \right) B_{-} e^{ka} \right]$$

$$A_{\pm} = \frac{e^{2ika}}{4} \left[\left(1 \mp \frac{ik}{k} \right) \left(1 + \frac{ik}{k} \right) e^{-2ka} + \left(1 \pm \frac{ik}{k} \right) \left(1 - \frac{ik}{k} \right) e^{2ka} \right] C$$

$$A_{\pm} = \frac{e^{2ika}}{4} \left[\left(2 + i \left(\frac{k}{k} \mp \frac{k}{k} \right) \right) e^{-2ka} + \left(2 - i \left(\frac{k}{k} \mp \frac{k}{k} \right) \right) e^{2ka} \right] C$$

$$T = \frac{|C|^2}{|A_{+}|^2} = \frac{1}{4}$$

$$|A_{+}|^2 = \left| \left[1 + \frac{i}{2} \left(\frac{k}{k} - \frac{k}{k} \right) \right] e^{-2ka} + \left[1 - \frac{i}{2} \left(\frac{k}{k} - \frac{k}{k} \right) \right] e^{2ka} \right|^2$$

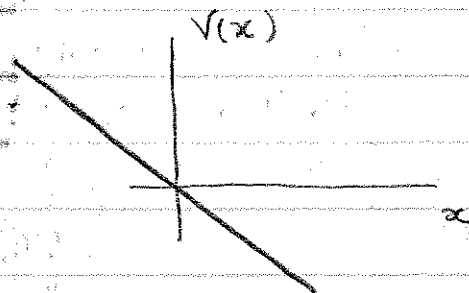
$$= \frac{1}{2}$$

$$\cosh 2ka + \frac{1}{4} \left(\frac{k}{k} - \frac{k}{k} \right)^2 \sinh 2ka$$

S5.4.3

$$U(p, t; p', 0) = \langle p | e^{-\frac{i}{\hbar} H t} | p' \rangle$$

$$H = \frac{p^2}{2m} - fx$$



$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - fx \right) \psi_E(x) = E \psi_E(x)$$

$$\left(+\frac{p^2}{2m} + \frac{\hbar}{i} f \frac{d}{dp} \right) \tilde{\psi}_E(p) = E \tilde{\psi}_E(p)$$

$$\frac{1}{\tilde{\psi}_E(p)} \left(\frac{\hbar}{i} f \frac{d}{dp} \right) \tilde{\psi}_E(p) = \frac{\hbar f}{i} \frac{d}{dp} \ln \tilde{\psi}_E(p)$$

$$= \left(E - \frac{p^2}{2m} \right) = \frac{d}{dp} \left(Ep - \frac{p^3}{6m} \right)$$

$$\tilde{\psi}_E(p) = N \exp \left[\frac{i}{\hbar f} \left(Ep - \frac{p^3}{6m} \right) \right]$$

$$= \langle p | E \rangle$$

$$\langle E' | E \rangle = |N|^2 \int_{-\infty}^{+\infty} dp e^{\frac{i}{\hbar f} (E - E') p}$$

$$= \delta(E - E') \quad \text{provided} \quad |N|^2 = \frac{1}{2\pi \hbar f}$$

with this normalization

$$\int_{-\infty}^{+\infty} dE |E\rangle \langle E| = 1$$

$$U(p, t; p', 0) = \int_{-\infty}^{+\infty} dE \langle p | E \rangle \langle E | p' \rangle e^{-\frac{i}{\hbar} E t}$$

$$= |N|^2 \int_{-\infty}^{+\infty} dE e^{\frac{i}{\hbar f} E (p - p' - \hbar f t)} = \frac{i}{\hbar f} \frac{p^3 - p'^3}{6m}$$

$$= |N|^2 2\pi \hbar f \delta(p - p' - \hbar f t) e^{\frac{i}{6m \hbar f} (p'^3 - p^3)}$$

$$= \delta(p - p' - \hbar f t) e^{\frac{i}{6m \hbar f} (p'^3 - p^3)}$$

$$U(x, t; x', 0) = \langle x | e^{-\frac{i}{\hbar} H t} | x' \rangle$$

$$= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' \langle x | p \rangle \langle p | e^{-\frac{i}{\hbar} H t} | p' \rangle \langle p' | x' \rangle$$

$$= \int_{-\infty}^{+\infty} dp \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} e^{\frac{i}{6m\hbar f} [(p-ft)^3 - p^3]} \cdot \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} (p-ft) x'}$$

$$= \int_{-\infty}^{+\infty} \frac{dp}{2\pi\hbar} e^{\frac{i}{6m\hbar f} [-3ft p^2 + 3(ft)^2 p - (ft)^3]} \cdot e^{\frac{i}{\hbar} (x-x') p} e^{-\frac{i}{\hbar} ft x'}$$

$$\frac{1}{2m} [-p^2 t + ft^2 p] + (x-x') p$$

$$= -\frac{t}{2m} \left[p^2 - \left(ft + \frac{2m}{t} (x-x') \right) p \right]$$

$$= -\frac{t}{2m} \left[p - \frac{1}{2} \left(ft + \frac{2m}{t} (x-x') \right) \right]^2 + \frac{t}{8m} \left(ft - \frac{2m}{t} (x-x') \right)^2$$

$$U(x, t; x', 0)$$

$$= \frac{1}{2\pi\hbar} \left[\pi \frac{2m\hbar}{it} \right]^{\frac{1}{2}} \exp \left\{ \frac{-i}{6m\hbar} f^2 t^3 \right. \\ \left. + \frac{i}{\hbar} \frac{t}{8m} \left(ft + \frac{2m}{t} (x-x') \right)^2 + \frac{i}{\hbar} ft x' \right\}$$

$$= \sqrt{\frac{m}{2\pi\hbar it}} \exp \left\{ \frac{i}{\hbar} \left[ft \frac{1}{2} (x+x') + \frac{m}{2t} (x-x')^2 \right. \right. \\ \left. \left. - \frac{1}{24m} f^2 t^3 \right] \right\}$$